

- Q.29 Define and classify the transducers. (CO5)
- Q.30 Explain the working of liquid thermocouple meter. (CO7)
- Q.31 Describe the various errors occur in induction type energy meter. (CO5)
- Q.32 Draw and explain the construction of frequency meter. (CO6)
- Q.33 Give the applications of torque tester. (CO4)
- Q.34 Draw the block diagram of digital Energy meter. (CO6)
- Q.35 Differentiate voltmeter and ammeter. (CO3)

#### SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the working principle and construction of a PMMC instrument. (CO2)
- Q.37 Explain the working principle and construction of a Dynamometer type watt meter. (CO2)
- Q.38 Explain the essentials of indicating instruments. (CO1)

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**4th Sem / Elect, Power Station Engg., Elect. & Eltx. Engg**  
**Subject:- Electrical Measuring Instruments and**  
**Instrumentation / Elect. & Eltx. Measuring Instr.**  
Time : 3Hrs.      M.M. : 100

#### SECTION-A

- Note:** Multiple choice questions. All questions are compulsory (10x1=10)
- Q.1 A CRO can be used to measure (CO1)
- Frequency
  - Power
  - Resistance
  - None of these
- Q.2 Two holes in the disc of energy meter are drilled at the opposite sides of the spindle to (CO2)
- Improve its ventilation
  - Eliminate creeping at no-load
  - Increase its deflecting torque
  - Increase its braking torque
- Q.3 Moving iron instruments can be used for (CO6)
- Both D.C and A.C
  - D.C only
  - A.C. only
  - None of these
- Q.4 To extend the range of voltmeter, a resistance is connected to it in (CO3)
- Series
  - Parallel with capacitor
  - Series-parallel
  - None of these

- Q.5 Thermocouples are (CO5)  
 a) Active transducer b) Passive transducer  
 c) Strain gauge d) None of these
- Q.6 LVDT is used to measure (CO5)  
 a) Temperature b) Velocity  
 c) Displacement d) Acceleration
- Q.7 RTD is used to measure (CO5)  
 a) Flow b) Pressure  
 c) Stress d) Temperature
- Q.8 Power in a three phase Unbalanced system can be measured by using : (CO6)  
 a) Single Watt meter method  
 b) By two watt meter method  
 c) Both (a) and (b)  
 d) None of these
- Q.9 Piezoelectric transducer consists of (CO5)  
 a) Copper wire b) Quartz crystal  
 c) Aluminum wire d) Silicon
- Q.10 Which type of instrument is prone to “Creeping” (CO5)  
 a) Voltmeter b) Energy meter  
 c) Ohm meter d) None of these

### SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 A wattmeter is an instrument that reads \_\_\_\_\_ consumed. (CO2)

- Q.12 An Ammeter is always connected in series.(T/F) (CO1)
- Q.13 Energy meter is an \_\_\_\_\_ type instrument. (CO1)
- Q.14 DMM stands for \_\_\_\_\_ (CO3)
- Q.15 A multimeter can measure \_\_\_\_\_ (CO3)
- Q.16 LVDT has two primary and one secondary.(T/F) (CO4)
- Q.17 CRT stands for \_\_\_\_\_ (CO2)
- Q.18 Thermoelectric transducers use \_\_\_\_\_ effect. (CO6)
- Q.19 VTVM stands for \_\_\_\_\_ (CO3)
- Q.20 Define Gross error. (CO1)

### SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the indicating and integrating instruments. (CO1)
- Q.22 Explain the different errors occurs in Moving coil instruments. (CO1)
- Q.23 Explain the principle of thermocouple. (CO5)
- Q.24 Describe the measurement of displacement with LVDT. (CO7)
- Q.25 Explain the construction and working of P.T. (CO4)
- Q.26 Explain the application of LCR meter. (CO6)
- Q.27 Draw the block diagram of CRO. (CO2)
- Q.28 Draw the circuit for measurement of power in a 3 phase system by using two watt meter method. (CO2)